

Article

Real-Time Strawberry Ripeness Classification and Counting: An Optimized YOLOv8s Framework with Class-Aware Multi-Object Tracking

Oluwasegun Moses Ogundele ¹, Niraj Tamrakar ¹, Jung-Hoo Kook ², Sang-Min Kim ², Jeong-In Choi ², Sijan Karki ¹, Timothy Denen Akpenpuun ³ and Hyeon Tae Kim ^{1,*}

¹ Department of Bio-Systems Engineering, Gyeongsang National University, (Institute of Smart Space Agriculture (ISSA)), Jinju 52828, Republic of Korea

² Department of Smart Farm, Gyeongsang National University, (Institute of Smart Space Agriculture (ISSA)), Jinju 52828, Republic of Korea

³ Department of Agricultural and Biosystems Engineering, University of Ilorin, PMB 1515, Ilorin 240103, Nigeria

* Correspondence: bioani@gnu.ac.kr; Tel.: +82-55-772-1896

Abstract

Accurate fruit counting is crucial for data-driven decision-making in modern precision agriculture. In strawberry cultivation, a labor-intensive sector, automated, scalable yield estimation is especially critical. However, dense foliage, variable lighting, visual ambiguity of ripeness stages, and fruit clustering pose significant challenges. To overcome these, we developed a real-time multi-stage framework for strawberry detection and counting by optimizing a YOLOv8s detector and integrating a class-aware tracking system. The detector was enhanced with a lightweight C3x module, an additional detection head for small objects, and the Wise-IOU (WIoU) loss function, thereby improving performance against occlusion. Our final model achieved a 92.5% mAP@0.5, outperforming the baseline while reducing the number of parameters by 27.9%. This detector was integrated with the ByteTrack multiple object tracking (MOT) algorithm. Our system enabled accurate, automated fruit counting in complex greenhouse environments. When validated on video data, results showed a strong correlation with ground-truth counts ($R^2 = 0.914$) and a low mean absolute percentage error (MAPE) of 9.52%. Counting accuracy was highest for ripe strawberries ($R^2 = 0.950$), confirming the value for harvest-ready estimation. This work delivers an efficient, accurate, and resource-conscious solution for automated yield monitoring in commercial strawberry production.

Keywords: multi-object tracking; occlusion; strawberry detection; YOLOv8s; yield estimation



Academic Editor: Aichen Wang

Received: 4 August 2025

Revised: 30 August 2025

Accepted: 2 September 2025

Published: 9 September 2025

Citation: Ogundele, O.M.; Tamrakar, N.; Kook, J.-H.; Kim, S.-M.; Choi, J.-I.; Karki, S.; Akpenpuun, T.D.; Kim, H.T. Real-Time Strawberry Ripeness Classification and Counting: An Optimized YOLOv8s Framework with Class-Aware Multi-Object Tracking. *Agriculture* **2025**, *15*, 1906. <https://doi.org/10.3390/agriculture15181906>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

The strawberry (*Fragaria × ananassa*), a globally consumed fruit recognized for its nutritional value, distinct flavor profile [1], and visual appeal [2,3], plays an essential economic role in many nations, including South Korea. In 2022, South Korea had a strawberry cultivation area of 5745 hectares, with a production volume of 158,807 tons [4]. It contributed USD 932 million to the domestic economy, which represents 14.7% of the nation's total vegetable production value [5]. However, this vital agricultural sector faces challenges as an aging farming workforce and the steady migration of young people to urban areas have created labor shortages and rising operational costs. These demographic

and financial pressures emphasize an urgent need for technological solutions that can reduce dependency on manual labor. Among these, accurate and timely yield estimation has become a crucial component that allows for better resource allocation optimization throughout the supply chain, including labor planning, transportation logistics, storage utilization, and marketing strategies. The highly perishable nature of strawberries further increases the need for precise forecasting, since errors can lead to post-harvest losses due to spoilage and mismatches between supply and demand. Automated systems offer a more reliable and efficient alternative to traditional methods, which often rely on rough empirical estimations or lengthy manual sampling, thereby providing more explicit guidance for planning harvest schedules and cultivation strategies.

In recent years, advances in computer vision and artificial intelligence have changed agricultural practices [6,7]. These technologies serve as the foundation for the automation of agronomic activities traditionally dependent on manual labor, with robotic fruit harvesting emerging as a key focus area [8,9]. Central to these autonomous systems is the machine vision module, responsible for fruit identification and localization. Earlier efforts in this area relied on traditional image processing techniques, using hand-crafted features such as color, shape, and texture to detect fruit under controlled conditions. For example, Zhang et al. [10] introduced an automated method to assess fruit ripeness and quality based on color variations. Their method involves generating a two-dimensional color histogram for each ripeness grade, then mapping the input fruit color onto a defined color index. In Wang et al. [11], a pixel thresholding method was applied to classify lychee fruits into predefined categories, while a geometric center-based matching technique was used to detect and distinguish between clustered fruits. Similarly, Karki et al. [12] employed different machine learning models to categorize strawberry ripeness stages using multiple color spaces (RGB, HLS, CIELAB, and YCbCr) alongside biometric features. Among the tested models, a feed-forward artificial neural network (ANN) utilizing the CIELAB color space had the highest classification accuracy. While these traditional approaches offer valuable foundations, computer vision algorithms that depend on feature extraction face significant limitations when dealing with complex, large-scale data [13]. Variations in lighting conditions, different growth orientations, and occlusions from leaves or nearby fruits present significant barriers to accurate detection. Strawberry ripeness classification is complicated due to simultaneous changes in color, shape, size, and texture across growth stages, attributes that conventional methods struggle to capture and analyze effectively.

The rapid advancement of deep learning, particularly Convolutional Neural Networks (CNNs), has shown remarkable potential in addressing these challenges. CNNs can autonomously extract hierarchical and discriminative features directly from image data, resulting in models with superior detection accuracy and better adaptability to in-field variations. Furthermore, the availability of flexible and well-documented training frameworks has eased their implementation, making them more practical for complex agricultural tasks. Within this paradigm, CNN-based Object detection models are broadly categorized into one-stage and two-stage networks. One-stage detectors, such as the You Only Look Once (YOLO) algorithm introduced by Redmon et al. [14] process the image in a single forward pass, making them faster than their two-stage detector counterparts, such as Faster Regional Convolutional Network (Faster R-CNN) [15], which generate candidate object regions first before refining the detection in a second stage. While two-stage detectors are known for higher accuracy, their computational complexity often limits their use in real-time field scenarios. Furthermore, continuous updates and iterations of the YOLO series have steadily improved the detection accuracy of one-stage networks [16], establishing the YOLO model series as a fundamental tool for real-time agricultural applications.

Many studies have utilized YOLO networks for strawberry detection [9,17,18]. However, despite this progress, two main challenges continue to hinder the transition from simple detection to reliable, automated yield estimation: accurately classifying fruit maturity and robustly handling occlusions. Throughout the growing season, strawberry plants simultaneously bear fruits at various stages of development, from unripe to ripe. For yield estimation to be accurate, it must consider ripeness. Simply counting visible fruits is insufficient; growers must distinguish between immediate, harvest-ready yield (ripe fruit) and potential future yield (unripe and semi-ripe fruit) to make informed management decisions. However, accurately classifying these ripeness stages under complex conditions, often complicated by severe occlusions, shadows, and inconsistent lighting, remains a significant research challenge. In facility agriculture, such as greenhouses, these complications are even more severe due to the varying sizes and cluttered, non-uniform distribution of strawberries that further hinders reliable detection and classification [19]. Therefore, improving the detection accuracy and performance of the base model is a crucial first step. This optimization must also ensure the architecture is streamlined with reduced complexity, making it suitable for deployment on resource-constrained edge devices. However, this is only part of the solution for a comprehensive yield analysis.

To perform an effective in-field count, this enhanced detection should serve as the foundation for a dynamic tracking and counting system capable of operating across video sequences to ensure each fruit is counted only once. Dynamic fruit counting in video sequences typically involves combining an object detector with an MOT algorithm. MOT algorithms leverage information from preceding and succeeding frames to establish object correspondence and predict a target's state, facilitating robust cross-frame association [20]. Traditional tracking-by-detection methods, such as SORT [21] and DeepSORT [22] utilize a Kalman filter for motion prediction and the Hungarian algorithm for data association, with DeepSORT adding a re-identification (Re-ID) model. More recently, ByteTrack [23] has demonstrated an excellent balance of speed and accuracy by retaining low-confidence detection boxes to effectively re-associate occluded targets without relying on computationally costly appearance features.

Although MOT research is thorough, it has mainly focused on domains like pedestrian tracking, intelligent driving, military aerospace, and security surveillance, where the targets are generally large with obvious characteristics [24–26]. The adaptation of these methods to agriculture presents unique obstacles related to visual similarities between fruits and occlusions. Nevertheless, MOT frameworks have been adapted for various yield estimation tasks in agriculture. For example, researchers have integrated YOLO with Kalman filters to count apples and camellia flowers [27,28] or combined it with DeepSORT for counting peanuts from UAV imagery [29]. However, these use cases mainly focus on crops that ripen uniformly. This approach is incompatible with strawberries, which exhibit continuous flowering and fruiting patterns, meaning multiple fruit stages can be present at the same time. In addition, in practical scenarios involving continuous camera motion relative to the strawberry fruit, as is common in greenhouse row-based data collection, counting all fruits detected throughout a video can lead to errors. Virtual line-crossing methods, proven in areas such as traffic counting, hold significant potential to improve fruit counting accuracy by providing a clear trigger for counting each fruit once as its trajectory intersects a defined boundary.

Table 1. Comparative analysis of recent studies on strawberry detection and counting.

Study	Detector Model	Tracker	Main Focus	Research Gap
Li et al. [30]	Improved Faster R-CNN	N/A (image-based)	High-accuracy detection and counting of ripe strawberries using a two-stage detector	Not suitable for real-time performance and focuses only on ripe fruits.
Zhou et al. [31]	YOLOv5s	Custom (MultiMap)	Designed an algorithm to handle severe occlusion in fruit clusters	Focuses only on occlusion and total strawberry counts.
Wang et al. [32]	DSE-YOLO	N/A (Detection only)	Enhanced detection using YOLO for improved detection of multi-stage strawberries	Focuses only on detection, not integrated for video-based strawberry counting.
Tamrakar et al. [33]	YOLOv5s-CGhostnet	N/A (Detection only)	Developed a lightweight detector for the strawberry ripeness level and counting	Lacks tracking and validation. The detector was not integrated into the end-to-end pipeline.

Despite the aforementioned advances, a detailed literature review reveals that most methods either focus on the optimization of the detection model in still images or address challenges such as occlusion without presenting a complete framework for counting that considers ripeness within video frames. Table 1 presents a comparative analysis of recent research works highlighting their main innovations and existing research gaps. As the comparison shows, significant gaps persist between high accuracy detection and the development of practical systems for strawberry counting. Studies like Li et al. [30] achieved high precision using an improved Faster RCNN, but the computational requirement makes it unsuitable for real-time analysis. While several other studies, such as those by Wang et al. and Tamrakar et al. [33] utilized the one-stage YOLO algorithm; however, the integration within a tracking and counting framework was not implemented, and the validation against manual ground truths was not conducted. Advanced tracking-focused works, such as those in Zhou et al. [31] concentrated on solving occlusion for total fruit count and did not consider the different ripeness levels critical for commercial harvesting decisions.

In response to these challenges, this study proposes an integrated, end-to-end framework specifically for strawberry ripeness classification and greenhouse in-field counting, which is built upon three key contributions. First, we present an optimized YOLOv8s detector that integrates a lightweight C3x module, an additional detection head tailored to improve strawberry detection due to their small size, and the Wise-IoU (WIoU) loss function to improve both efficiency and robustness against occlusion. Second, we detail the integration of the improved detector with the ByteTrack algorithm by implementing modifications to its Kalman filter, which enables accurate distinction between ripeness stages, a vital requirement for actionable yield data. Third, we incorporate these components into a robust counting methodology using a virtual line-crossing method with a precisely defined ROI to improve counting accuracy as the camera moves along the greenhouse cultivation rows. Finally, the performance of this complete framework is validated using a comprehensive custom dataset from a greenhouse environment. The proposed framework aims to provide theoretical support and a practical solution for automated yield estimation, facilitating timely and informed decision-making for strawberry growers.

2. Materials and Methods

2.1. Methodology

The methodology for this study uses a structured multi-stage framework to achieve improved strawberry ripeness classification for greenhouse-based fruit counting applications. The first step involved the collection of two datasets: an image dataset for training and testing strawberry detection models, and a video dataset for developing the counting algorithm. We developed an improved YOLOv8s specifically for strawberry detection and classification tasks. After achieving good performance compared to other algorithms, we incorporated the improved model as the main detector within an automated counting system. This system processes video streams and uses the ByteTrack MOT algorithm to accurately identify, track, and count individual strawberries according to the ripeness class across frames. We validated the counting performance through regression analysis. The complete methodology flowchart is shown in Figure 1.

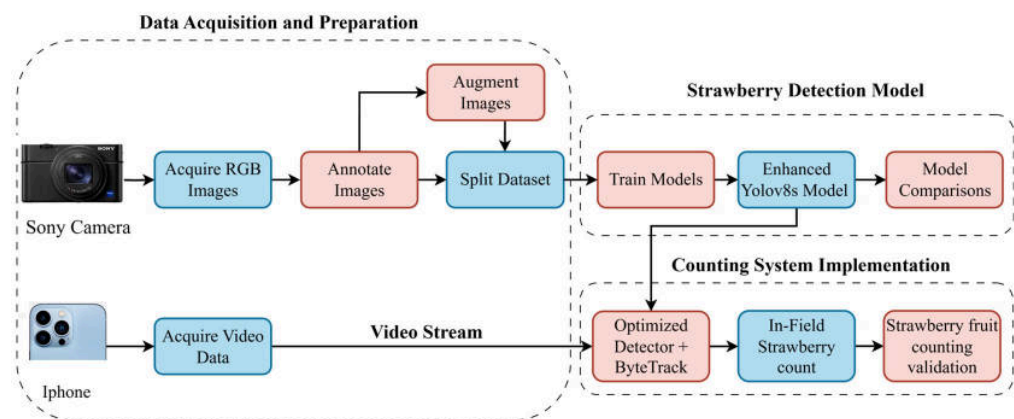


Figure 1. Flowchart of the study.

2.2. Strawberry Dataset

This study utilized a custom strawberry dataset collected from a greenhouse facility managed by the Smart Space Sensing Laboratory (SSSL), Gyeongsang National University, located in Jinju, Gyeongsangnam-do (South Gyeongsang Province), South Korea, at latitude 35.09° N and longitude 128.05° E. The dataset included images and videos of three popular Korean strawberry cultivars: *Seolhyang*, *Keumstil*, and *Honghee*. All these varieties belong to the hybrid species *Fragaria* × *ananassa*. These cultivars were grown on five raised cultivation beds from September to February for three consecutive growing seasons: 2022/2023, 2023/2024, and 2024/2025. Data acquisition was conducted during the principal harvesting periods, December and February of each season, to account for the phenotypic variability associated with strawberry maturity.

RGB images and videos of strawberries were captured using two devices: the Sony Cyber-shot DSC-RX100 VII (specifications provided in Table 2) for still images, and a high-resolution mobile camera (Apple iPhone 13 Pro Max) for video acquisition. To maintain consistency and reproducibility during the collection process, the devices were mounted on a mobile platform at a height of 0.8 m, capturing fruits with distances ranging from 30 to 80 cm. It was collected under varying natural lighting conditions and at different periods during the day: morning (around 9 a.m.), afternoon (around noon), and evening (around 5 p.m.), as depicted in Figure 2. This temporal and lighting variability was intentionally introduced to enhance the generalizability and robustness of the deep learning model, as model performance typically improves with diversity of training data.

Table 2. Specification of Sony Cyber-shot DSC-RX100 VII.

Specifications	Details
Manufacturer and model	Sony Cyber-shot DSC-RX100 VII
Image sensor	1.0-type stacked (13.2 mm × 8.8 mm) CMOS
Effective resolution	Approximately 20.1 megapixels
Color stream format	RGB8
White balance and exposure	Automatic



(a) Morning



(b) Noon



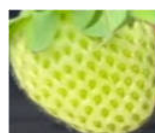
(c) Evening

Figure 2. Examples of strawberry images captured under varying natural lighting conditions and at different times of the day.

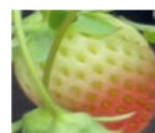
The ripeness level of the strawberry images was divided into three distinct categories: Category I (unripe), Category II (semi-ripe), and Category III (ripe). The classification was based on the amount of surface redness and used a protocol similar to that of our previous study [34]. The ripeness levels are summarized in Table 3. Additionally, the dataset included strawberries from all three ripeness levels (Category I, II, III) with varying sizes taken in naturally complex greenhouse conditions, as illustrated in Figure 3. These conditions included dense clusters of fruits, as well as partial occlusion by leaves, overlapping fruits, and slight changes in camera angles and fruit orientation. Capturing the different, challenging instances was essential for improving the model's robustness and preventing overfitting. This approach enabled the detection system to learn under conditions that closely resemble real-world agricultural environments.

Table 3. Classification of strawberry ripeness levels based on external appearance.

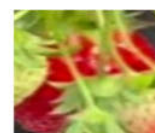
S/No.	Category	Appearance
1	I	Predominantly greenish/whitish
2	II	Approximately 75% or less of the surface is red
3	III	Fully red surface



(a)



(b)



(c)



(d)



(e)



(f)



(g)



(h)

Figure 3. Sample of strawberries under varied field conditions: (a) Category III with clear visibility; (b) Category II under soft occlusion; (c) Dense fruit cluster with mixed categories; (d) Partial leaf occlusion affecting two categories; (e) Category I with strong visibility; (f) Category II under moderate blur; (g) Side-angled Category II; (h) Overlapping Category I and III.

2.3. Dataset Construction and Preprocessing

A total of 2190 strawberry images and 37 videos were collected from the greenhouse environment. The image dataset for training and evaluating the detection model was processed first. The Label studio (version 1.16.0) annotation software was used to manually label all 2190 images, with bounding boxes drawn around each strawberry according to the ripeness categories (see Figure 4 for interface details). The dataset had 13,228 annotated instances, with an average of approximately 6 fruits per image. The distribution included Category I, II, and III, which had 6133, 1675, and 5420 instances, respectively.

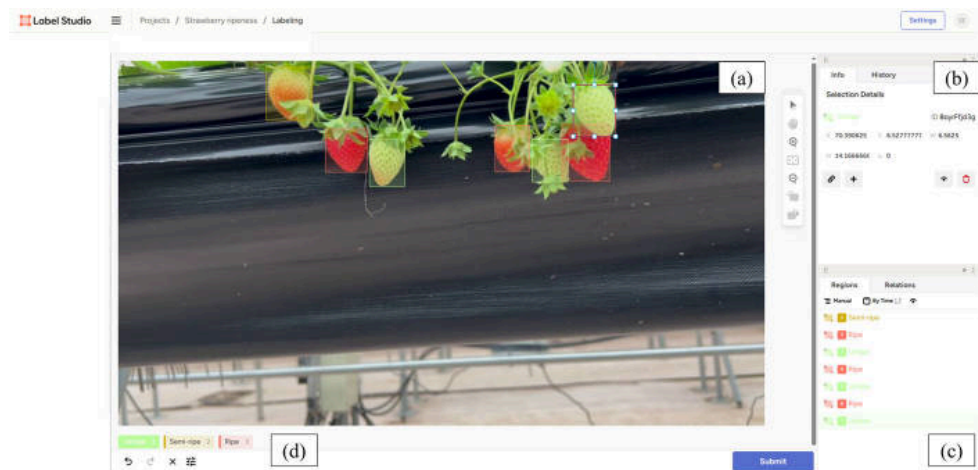


Figure 4. Label studio software used for annotation: (a) Labeling window; (b) Current selection details; (c) List of classes annotated; (d) Total class list.

For the subsequent task of validating the counting algorithm, we selected a subset of 20 videos from the 37 initially collected. The selection criteria prioritized videos exhibiting more stable camera motion to ensure a reliable basis for ground-truth. For this validation set, the ground truth was established by two independent operators manually counting all strawberry fruits of each ripeness level in every row after data collection. While minor discrepancies seldom occurred in highly occluded scenes or under low-light conditions, the inter-operator error margin was minimal at an average of approximately 2 fruits per video, indicating the reliability of the manual counts for validation purposes.

To improve the diversity of the training dataset, three augmentation techniques, namely rotation, Gaussian blur, and random brightness and contrast, were used. Rotation involved small-angle shifts to simulate natural variation in fruit orientation. Gaussian blur was introduced using a low-kernel filter to mimic motion or focus imperfections. Random brightness and contrast adjustments were then applied to make the model more robust to the diverse lighting conditions encountered in the field, such as variations due to time of day, cloud cover, and shadows from foliage.

The final dataset contained 4250 images, and the annotations were exported in the YOLO format, with each file containing bounding box information that details the class label and normalized coordinates. This dataset was partitioned into training, validation, and testing sets with a ratio of 8:1:1 for model development and evaluation. The complete data and preprocessing workflow is shown in Figure 5.

2.4. YOLOv8 Architecture

The YOLOv8 model, developed by Ultralytics in 2023, marks a significant evolution in the You Only Look Once (YOLO) family of real-time object detection models. It has become increasingly popular in the domain of agricultural computer vision applications [35–37]. The YOLOv8 enhances the capabilities of earlier versions by combining object detection,

classification, and segmentation into a single unified framework optimized for real-time performance. Unlike earlier models that relied on anchor boxes, YOLOv8 adopts an anchor-free detection approach, making predictions directly at the center of objects. This eliminates manual anchor tuning and accelerates Non-Maximum Suppression (NMS), which improves accuracy, particularly for small object detection. It also uses mosaic augmentation that composites four training images into synthetic mosaics to improve model robustness to occlusion and variations in scale. Together, these features improve YOLOv8's precision, speed, and flexibility, allowing for high-performance inference even on devices with limited resources [38].

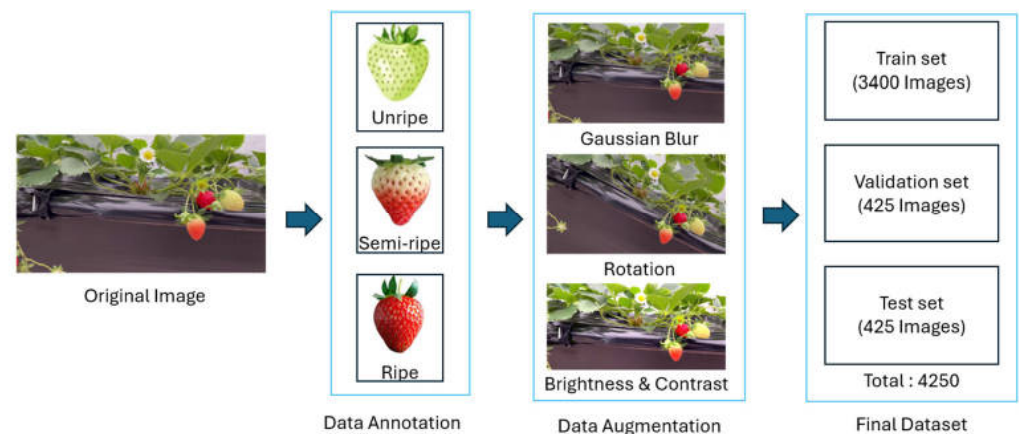


Figure 5. Data collection and preprocessing workflow.

The YOLOv8 architecture, shown in Figure 6, has four main components: input, backbone, neck, and head. The input module manages image normalization and pre-processing. The backbone, based on the CSPDarknet structure [39], includes three key modules: the Conv, C2f, and SPPF (Spatial Pyramid Pooling Fast) layers. The Conv blocks extract essential features and utilize the SiLU activation function and batch normalization to normalize feature representations. The C2f module improves feature representation through cross-layer fusion. The SPPF layer gathers multi-scale spatial features, thereby enhancing robustness in detecting objects of different sizes. These features are passed to the neck, which employs a PANet-FPN architecture for multi-scale feature fusion [40]. The neck contains the Upsample, Concat, and C2f modules. The Upsample layer increases low-resolution feature maps to match high-resolution ones. In contrast, the Concat layer combines features from various levels, enabling the model to detect large, medium, and small objects simultaneously. The C2f module further integrates multi-scale information to ensure feature maps carry rich contextual data, ultimately increasing detection performance. The head has a decoupled structure that separates classification and bounding box regression tasks into parallel branches. This design resolves the inherent conflict between the 'what' (classification) and 'where' (regression) objectives, resulting in improved convergence and higher accuracy. It comprises a series of C2f and convolutional layers that process the multi-scale feature maps produced by the neck. These processed features are then sent to the final Detect layer, which generates prediction vectors containing class probabilities and bounding box coordinates for all potential objects.

While computer vision algorithms like YOLO are foundational for agricultural automation, from robotic harvesting to yield estimation, their generalization capabilities, though well-established on general-purpose datasets like COCO [41], have not been extensively verified on specialized agricultural datasets for tasks such as strawberry detection. This limitation arises from the unstructured nature associated with the strawberry growing environment, characterized by highly variable lighting conditions, severe occlusion from

leaves and stems, and morphological variations among fruit cultivars. This study focuses on improving YOLOv8 for real-time in-field strawberry detection and counting. Therefore, it is essential to select a model variant that strikes an optimal balance between high detection accuracy and low inference time. This is essential for potential deployment in resource-constrained environments such as greenhouses.

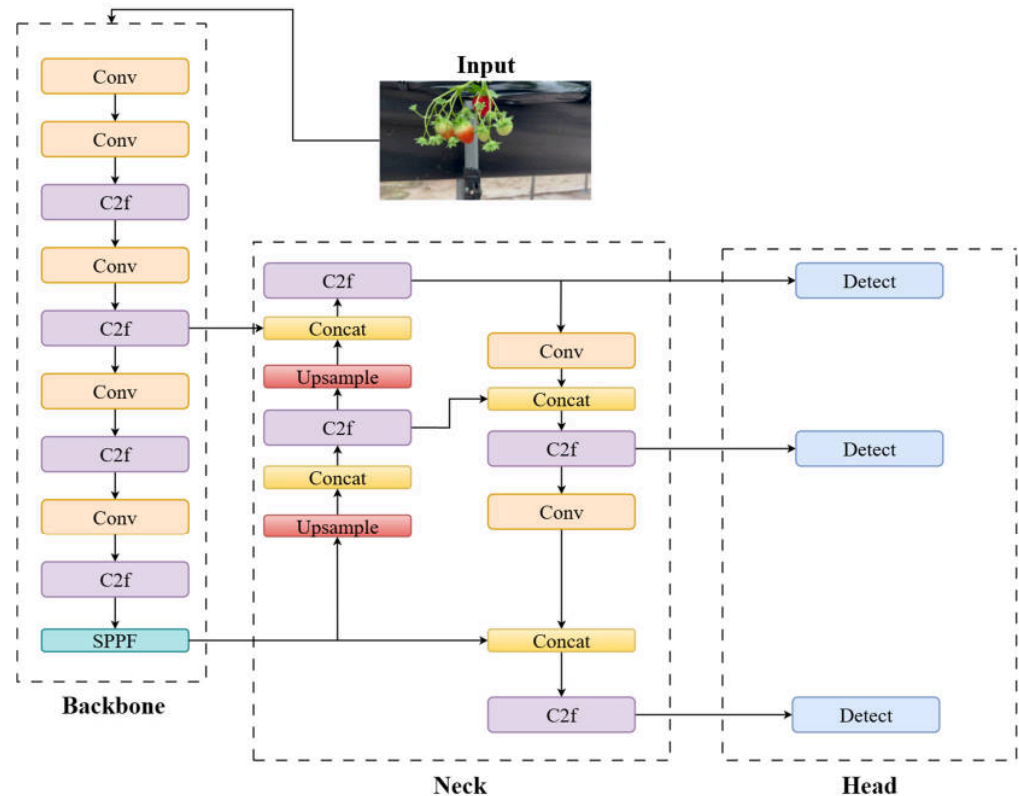


Figure 6. YOLOv8 network architecture. The image illustrates the four main sections (input, backbone, neck, and head).

The YOLOv8 model comes in five variants: nano (YOLOv8n), small (YOLOv8s), medium (YOLOv8m), large (YOLOv8l), and extra-large (YOLOv8x). These variants differ in model depth, width, and maximum number of channels, which collectively affect detection accuracy, inference speed, and model size. YOLOv8s represents a well-balanced compromise between speed and performance, as reported in similar recent research studies [42,43]. Therefore, the YOLOv8s model was chosen as the baseline for this study.

2.5. Improved YOLOv8 Architecture

2.5.1. Module and Head

The small size of strawberries, their visual similarities, and complex growing environment, which includes overlaps and occlusions, necessitate target optimization of the original YOLOv8 framework for this application. Since the classification task in this study involves only three classes: Category I (unripe), Category II (semi-ripe), and Category III (ripe). The standard YOLOv8 framework, which is optimized for the diverse and multi-class COCO dataset, has a higher complexity than required for the targeted three-class detection task. Our primary modifications, therefore, target the C2f modules within the model's backbone and the detection head, which are major contributors to the model's parameter count and detection accuracy.

As a result, we made two adjustments. Firstly, we replaced the standard C2f modules, shown in Figure 7a, with the more computationally efficient C3x modules, shown in

Figure 7b. Its key features include: (i) it is based on the proven C3 module from YOLOv5 but incorporates a more efficient cross-convolution mechanism. (ii) The module splits the input feature map into two parallel branches. The channel count is halved in each branch using a Conv operation before the resulting feature maps are concatenated, effectively merging information. (iii) The core innovation lies in its bottleneck design, which employs sequential 1×3 and 3×1 convolutional layers. This asymmetric structure significantly reduces the parameter count and computational load compared to the standard square kernels used in C2f, while still effectively capturing spatial features and merging channel information. A comparison between the two modules on our strawberry dataset is shown in Table 4.

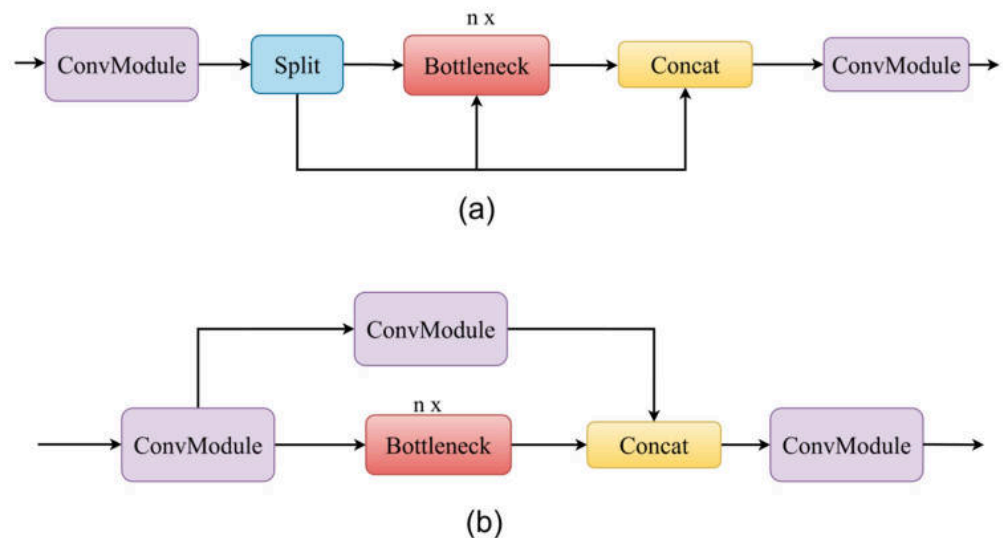


Figure 7. Module architecture (a) C2f module; (b) C3x module.

Table 4. Comparison between the two modules (C2f and C3x).

Module	Parameter Size	Kernel Size
C2f	11126745	[3 3] [3 3]
C3x	8607705	[3 1] [1 3]

Secondly, for enhancing detection accuracy, particularly for the small fruit characteristic of ripeness Category I (unripe) and II (semi-ripe), a second modification was made. The standard YOLOv8 architecture can struggle with such targets due to the loss of fine-grained spatial information during successive down-sampling. To solve this, the detection head was enhanced by introducing an additional small target detection layer. This new layer is specifically designed to operate on a higher-resolution feature map (160×160) provided by the neck network. By processing features at this finer scale, the model effectively retains the detailed spatial information necessary for localizing small objects that are often indistinct or lost in the standard, lower-resolution detection heads, thereby significantly improving performance on the most challenging targets within the canopy. The improved model architecture is presented in Figure 8.

2.5.2. Loss Function

The intersection-over-union (IoU) loss function is an essential criterion for assessing the performance of object detection models [44]. IoU compares the predicted labels with the ground truth labels. It plays a vital role in determining both negative and positive labels, as well as measuring the proximity between predicted and true labels. In Figure 9,

the blue rectangle represents the ground truth box, and the yellow rectangle represents the anchor box.

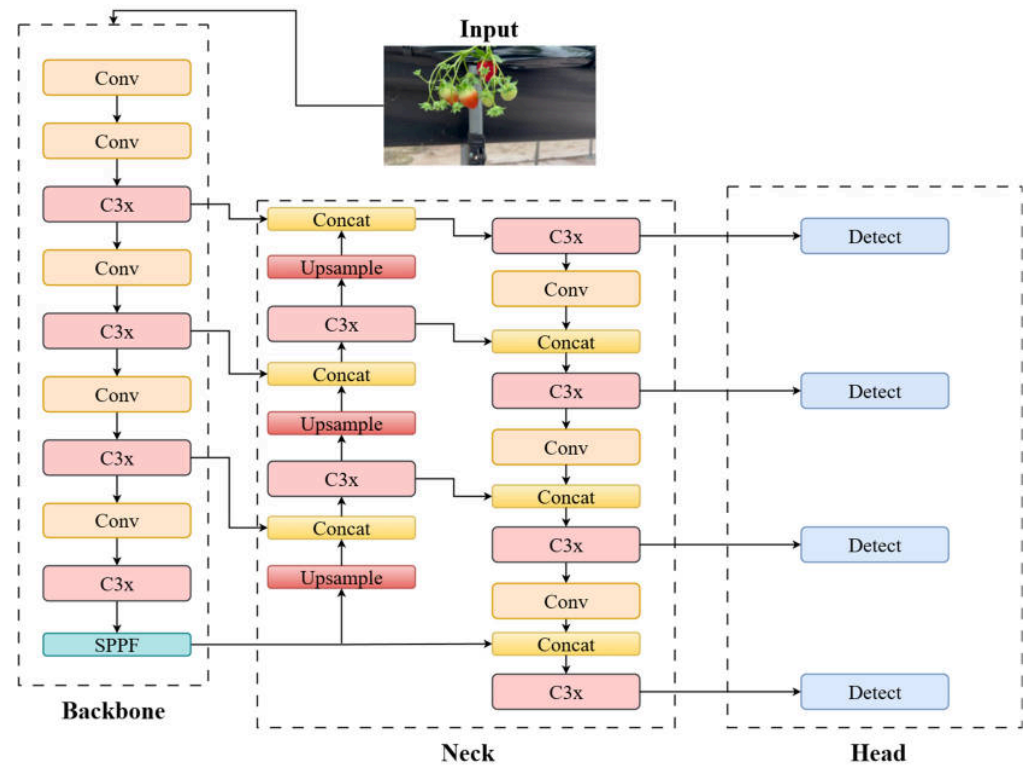


Figure 8. Improved YOLOv8 architecture, including C3x module and detection head.

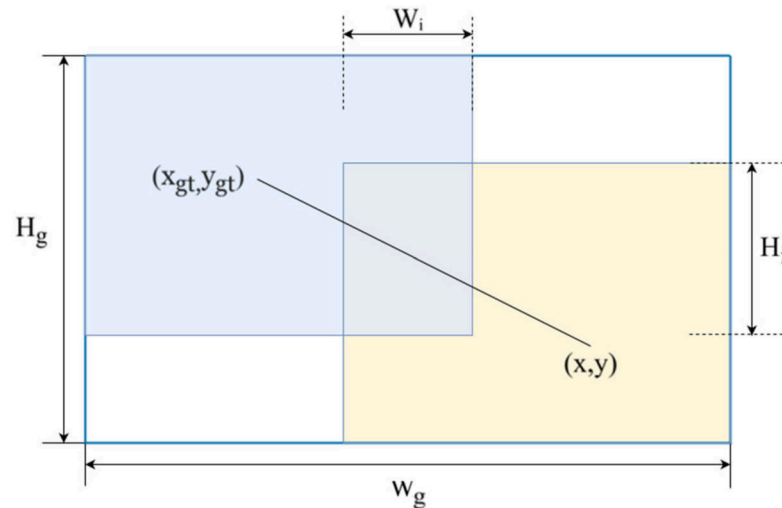


Figure 9. Schematic diagram of the anchor box and ground truth box.

In YOLOv8, the complete intersection-over-union (CIoU) loss function (Equation (4)) [45] is employed, which considers the overlapping area, center point, and aspect ratio of the bounding box regression. Equations (1)–(3) are used to determine the Complete IoU regularization term (R_{CIoU}), aspect ratio weighting coefficient (α), and aspect ratio consistency term (v).

$$R_{CIoU} = \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (1)$$

$$\alpha = \frac{v}{(1 - IoU) + v} \quad (2)$$

$$v = \frac{4}{\pi^2} - \left(\left(\arctan \frac{w^{gt}}{h^{gt}} \right) - \arctan \frac{w}{h} \right)^2 \quad (3)$$

$$L_{CIoU} = 1 - L_{IoU} + R_{CIoU} \quad (4)$$

where b and b^{gt} represent the centroids of the predicted anchor box and the ground truth box, respectively; $\rho(b, b^{gt})$ refers to the Euclidean distance between these centroids. c denotes the diagonal length of the smallest bounding box that contains both the anchor and ground truth boxes. α is a balancing coefficient, and v quantifies the consistency between the aspect ratios of the two boxes. w^{gt} and h^{gt} indicate the width and height of the ground truth box, while w and h correspond to the width and height of the predicted anchor box.

While the CIoU loss function improves upon standard IoU by incorporating overlap area, centroid distance, and aspect ratio, CIoU still has inherent limitations. In challenging detection scenarios such as our strawberry dataset involving small, low-quality, or poorly positioned objects, CIoU tends to harshly penalize low-quality anchor boxes due to its sensitivity to spatial resolution and object aspect ratio. This can result in degraded generalization performance, unstable gradients during training, slower convergence, and suboptimal localization.

To address these shortcomings and accelerate convergence, this study adopts the Wise IoU (WIoU) loss function as proposed by Tong et al. [46]. Unlike CIoU, WIoU introduces the concept of “outlier degree” to dynamically assess anchor quality, enabling a balanced gradient distribution. By reducing the influence of low-quality samples while focusing on informative high-quality anchors, WIoU enhances the model’s robustness and localization precision. This dynamic non-monotonic adjustment helps the network better differentiate between difficult, complex, and easy samples, making WIoU a more suitable loss function for complex, small-object detection tasks like strawberry detection. There are three versions of the WIoU loss function: WIoU v1 (Equation (5)), v2, and v3 (Equation (8)). We have selected WIoU v3, which was constructed from WIoU v1 (See Equations (5)–(10)). It extends the capabilities of WIoU v1 by introducing a non-monotonic focusing factor r (Equation (9)). This dynamically optimizes the weights in the loss function, reduces harmful gradients from low-quality samples, and allows for hyperparameters α to be tuned to suit different models. This dynamically optimizes the weights in the loss function, reduces harmful gradients from low-quality samples, and allows for hyperparameters α to be tuned to suit different models.

$$L_{WIoUv1} = R_{WIoU} L_{IoU} \quad (5)$$

$$R_{WIoU} = \exp\left(\frac{(x - x_{gt})^2 + (y - y_{gt})^2}{(W_g^2 + H_g^2)}\right) \quad (6)$$

$$L_{IoU} = 1 - IoU \quad (7)$$

$$L_{WIoUv3} = r L_{WIoUv1} \quad (8)$$

$$r = \frac{\beta}{\delta \alpha^{\beta - \alpha}} \quad (9)$$

$$\beta = \frac{L_{IoU}^*}{L_{IoU}} \in [0, +\infty) \quad (10)$$

where W_g and H_g represent the width and height of the smallest outer rectangle of the anchor box and the ground truth box, (x, y) , and (x_{gt}, y_{gt}) denote the center coordinates of the anchor box and the ground truth box, respectively. L_{IoU}^* is the monotonic focusing factor, while β is the outlier degree of the anchor box.

2.6. ByteTrack Algorithm

The transition from single-image detection to dynamic, in-field yield estimation through fruit counting was achieved by developing a counting system based on an MOT framework. The core of this system integrates our previously described, improved YOLOv8s model with the ByteTrack algorithm, as shown in Figure 10 [47].

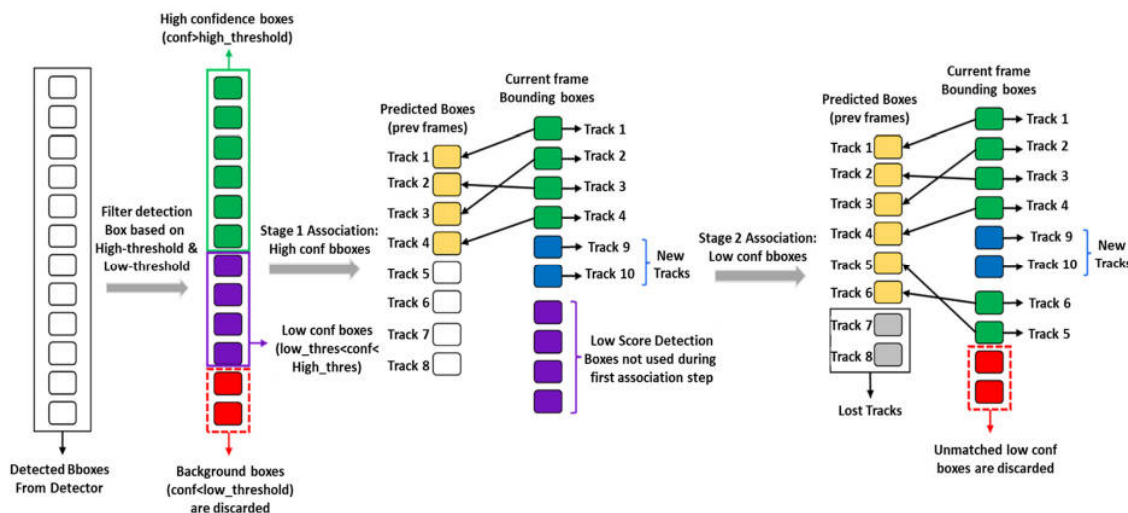


Figure 10. ByteTrack algorithm.

To address the significant challenges in tracking dense, small objects like strawberries within a greenhouse, such as frequent occlusions by leaves, overlapping fruits, and potential ID-switching because of camera motion, we selected ByteTrack. Unlike traditional MOT methods that often discard detection boxes with low confidence scores, ByteTrack employs an innovative data association strategy called BYTE. The core principle of BYTE is to retain nearly all detection boxes, separating them into two categories: high-confidence and low-confidence. The tracking process happens in two stages:

(i) high-confidence detections are first matched with existing tracklets using a Kalman filter to predict their new locations. (ii) low-confidence detections are then used to match with the remaining unmatched tracklets. This step is key for recovering objects that are temporarily occluded, as their detection confidence may briefly drop.

To tailor this framework specifically for strawberry ripeness counting, we introduced a critical enhancement for multi-class state estimation. To maintain distinct counts for each ripeness category, we integrated class information directly into the tracking state. The standard Kalman filter state vector was augmented to include a class variable, class, as shown below:

$$x = [u, v, s, r, \dot{u}, \dot{v}, \dot{s}, class]^T \quad (11)$$

where (u, v) denotes the center coordinates of the bounding box, s is the aspect ratio, r is the height, and $(\dot{u}, \dot{v}, \dot{s})$ are their respective velocities.

By including the class variable, the matching process between predicted and detected objects is performed on a per-class basis, preventing a Category III (ripe) strawberry from being mismatched with a Category I (unripe) one.

Finally, to convert the continuous tracking data into a discrete count, a virtual line counting methodology was used, anchored by a precisely defined Region of Interest (ROI) within the 1920×1080 -pixel video frame (Figure 11). This ROI was configured as a vertically oriented counting zone, at 500×1080 pixels, and was horizontally centered. The primary trigger for counting was the entry of a tracked strawberry's bounding box centroid into this ROI. To ensure each fruit was counted only once, the system performed

a uniqueness check upon entry of the ROI. The track-ID of the entering fruit was cross-referenced against a registration set of all previously counted IDs. If the ID was new, it was added to the set, and the count for its specific ripeness class was incremented. If the ID was already registered, it was ignored, thereby preventing recounting due to occlusion or erratic movement. This robust method guarantees high-fidelity enumeration by transforming continuous tracking data into a reliable, discrete count.

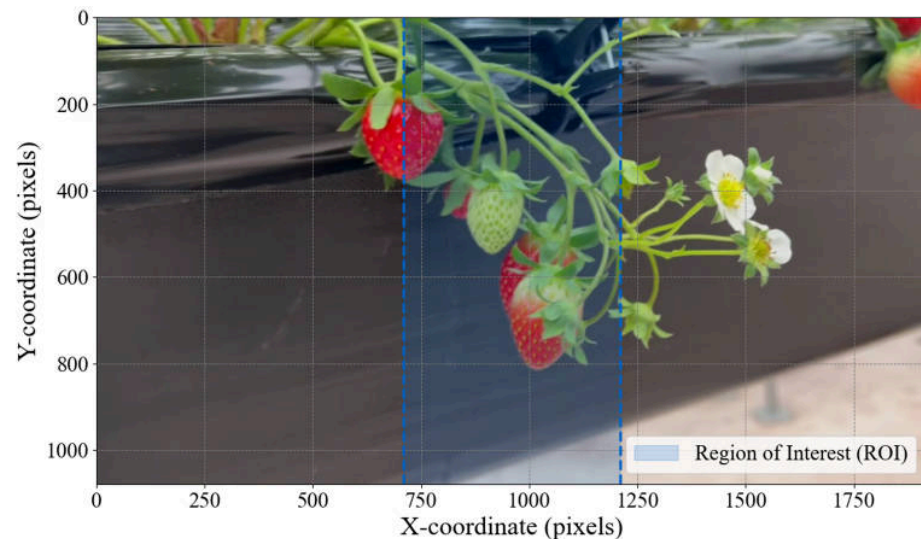


Figure 11. The virtual ROI for strawberry counting.

2.7. Experimental Platform and Related Settings

The test platform (hardware and software) for training and evaluating the deep learning models is presented in Table 5.

Table 5. Experimental testing platform.

Hardware/Software Environment	Parameter	Version
CPU	Intel(R) Core(TM) i7-10700	Frequency: 2.90 GHz
GPU	NVIDIA GeForce RTX 2060	Video Memory: 6 GB
Computer system	Windows 10 Pro	RAM: 16 GB
Deep learning framework	Pytorch	2.4.1
Computer platform	CUDA	12.1
Programming language	Python	3.8.2

2.8. Evaluation Metrics

2.8.1. Object Detection Model

This study utilizes precision, recall, F1 score, and mean average precision at 0.5 intersection over union (mAP@0.5) to quantitatively assess the performance of the proposed strawberry detection and classification models. The evaluation process is predicated on the Intersection Over Union (IoU), which measures the geometric overlap between a model's predicted bounding box and the ground truth. For this study, a detection was considered a True Positive (TP) if the IOU was 0.5 or greater and the predicted class label was correct. The evaluated metrics were calculated as:

Precision (P), also known as positive predictive value, measures the accuracy of the model's optimistic predictions. It is defined as the ratio of correctly identified strawberries

(TP) to the total number of predicted instances (both true positives and false positives, FP). A high precision indicates a low false positive rate.

$$P = \frac{TP}{TP + FP} \quad (12)$$

Recall (R), also known as sensitivity, measures the model's ability to identify all actual strawberries present in the images. It is the ratio of correctly identified strawberries (TP) to the total number of actual ground-truth instances (the sum of true positives and false negatives, FN). A high recall indicates a low false negative rate.

$$R = \frac{TP}{TP + FN} \quad (13)$$

F1-Score is used to provide a single, balanced measure of a model's performance. It is calculated as the harmonic mean of Precision and Recall, which is more robust than a simple average, especially in cases of class imbalance.

$$F1\ score = 2 \times \frac{P \times R}{P + R} \quad (14)$$

Average Precision (AP) and mean Average Precision (mAP): To evaluate performance across different confidence thresholds, the Precision-Recall (PR) curve is generated for each class. The AP for a single class is the area under its PR curve, providing a clear summary of the model's performance for that class. The primary evaluation metric for this study is the mAP, which is the mean of the AP values calculated across all ripeness categories ($n = 3$ in our case). We report mAP@0.5, which indicates that the mAP is computed using the IoU threshold of 0.5.

$$AP = \int_0^1 P(R) dR \quad (15)$$

$$mAP@0.5 = \frac{1}{N} \sum_{i=1}^N AP_i^{@0.5} \quad (16)$$

where $P(R)$ is the precision-recall curve, N is the total number of classes, and $AP_i^{@0.5}$ is the average precision for class i computed with an IoU threshold of 0.5.

2.8.2. Counting Algorithm

The accuracy of the automated counting algorithm was validated against manual ground-truth counts collected in the greenhouse. A linear regression analysis was used to evaluate the relationship between the algorithm's predicted counts and the actual counts. As part of the evaluation, the coefficient of determination (R^2) was calculated to measure the goodness of fit between the predicted and actual values, indicating how effectively the model explains the observed variation in the data.

In addition to R^2 , three standard error metrics were employed to quantify the magnitude of the counting errors: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). These metrics offer insights into prediction accuracy by capturing both absolute and relative error magnitudes. The equations used are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (17)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (18)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (19)$$

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (20)$$

where $\hat{y}_i$ is the predicted count, y_i denotes the ground-truth count, $\bar{y}$ is the mean of the ground-truth counts, and n is the total number of samples.

3. Results and Discussion

3.1. Comparative Performance of Detection Models

The evaluation of our improved YOLOv8s model (C3x module + Head + WIoU) was performed through a comprehensive benchmark against other mainstream YOLO algorithms, including YOLOv3-tiny, YOLOv5s-p6, YOLOv6s, YOLOv8n, YOLOv8s, and YOLOv8m. Before presenting the comparative benchmarks, we visualized the standalone performance of our improved model. As illustrated in Figure 12, the model effectively detects and classifies strawberries across all three ripeness stages, under the inherent complexities of the greenhouse environment. For example, the model successfully identifies multiple, overlapping Category III (ripe) fruits within dense clusters in Figure 12c,f. In Figure 12a,b,d and e, it correctly distinguishes between adjacent and clustered Category I (unripe), Category II (semi-ripe), and Category III (ripe) strawberries. It also maintains high accuracy on small, unripe fruits that are partially occluded by foliage in Figure 12d, highlighting the effectiveness of the architectural improvements.

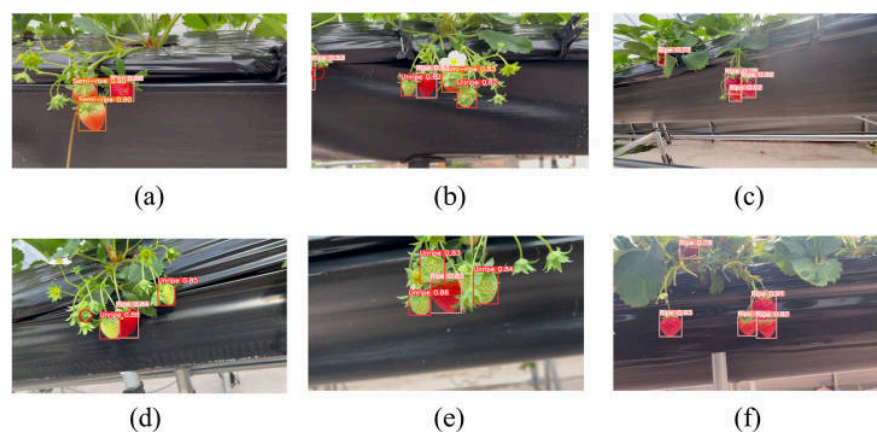


Figure 12. Visualization of the proposed model's detection performance under diverse and challenging in-field conditions. The sub-panels (a–f) illustrate successful localization and classification across all three ripeness stages, including limitations marked in red circles.

However, despite this high overall performance, the complex canopy environment still leads to some detection errors. For instance, in Figure 12d, the model correctly identifies a small unripe strawberry but misses another tiny, unripe fruit (marked by the red circle), highly occluded by a leaf, an example of a false negative caused by severe occlusion. Furthermore, another observable limitation appeared with strawberries positioned at the image edge. As shown by the red circle in Figure 12b, a ripe strawberry at the far-left edge of the frame was detected with a low detection certainty. In addition, the predicted bounding box for this instance was poorly localized. This phenomenon is characteristic of challenges posed by edge-of-frame objects, since the convolutional filters can only process partial information from these targets, resulting in weaker feature representation.

A detailed quantitative analysis, with results compiled in Table 6, confirms the superiority of our proposed model. The primary indicator of overall model performance,

mAP@0.5, was 92.5%. This represents a consistent improvement over YOLOv3-tiny, YOLOv5-p6, YOLOv6s, YOLOv8n and YOLOv8s with performance gains of 2.4%, 1.5%, 2.0%, 2.0% and 0.4%, respectively. The larger YOLOv8m achieved the highest detection accuracy of 92.6% mAP, a marginal 0.1% improvement. However, as will be explained in Section 3.4, this minor gain in performance comes at a significant computational cost, making it less suitable for our target application. Our model achieved a recall of 89.6%, a substantial increase of 3.4% over the baseline YOLOv8s and other models. This significant increase in recall shows an improved ability to identify all true strawberries in a frame. It reduces the missed detections. This balance between precision and recall is captured by the F1-score, which depicts a more comprehensive measure of a model's performance. Our model achieved the highest F1-score of 88.3%, outperforming the baseline YOLOv8s of 86.5% and demonstrating a more reliable and well-rounded detection ability.

Table 6. Comparative summary of the results of the different algorithms.

Algorithm	P (%)	R (%)	F1 Score	mAP@50 (%)
YOLOv3 tiny	84.4	86.2	85.3	90.1
YOLOv5s-p6	84.6	86.9	85.7	91.0
YOLOv6s	85.0	87.2	86.1	90.5
YOLOv8n	81.7	88.7	85.1	90.5
YOLOv8s	86.7	86.2	86.5	92.1
YOLOv8m	86.4	89.3	87.8	92.6
Ours	87.0	89.6	88.3	92.5

To better interpret these metrics, a visual comparison of our model against YOLOv3-tiny, YOLOv5-p6, YOLOv6s, and YOLOv8s is presented in Figure 13. This assessment serves to corroborate the quantitative data. Our improved model showed strong detection capabilities with high accuracy across all three ripeness categories. Meanwhile, other models showed notable limitations. As shown by the red for missed detections and yellow circles for misclassifications, YOLOv3-tiny and YOLOv5-p6 were particularly prone to false negatives and misclassifications. This directly relates to their lower recall and precision scores. While the performance of the baseline YOLOv8s and YOLOv6s performed well, both showed a significant drop in performance when detecting partially occluded strawberries. This visual analysis confirms that the improvements made to our model provide benefits in the most difficult detection scenarios, validating the practical impact of our optimization strategy.

3.2. Analysis of Per-Class Detection Accuracy

The primary objective of the detection and classification algorithm is to identify the location and class of strawberries for in-field counting. While a high overall mAP is essential. A more detailed, per-class evaluation is crucial to understand the model's strengths and limitations and to diagnose potential sources of error in the counting pipeline. Therefore, the performance of our improved YOLOv8s model was evaluated for each of the three ripeness classes. The per-class precision, recall, and AP@0.5, detailed in Table 7, revealed a nuanced performance profile that is directly linked to the visual characteristics of each growth stage.

Table 7. Class-wise performance of our improved model.

Ripeness Category	P (%)	R (%)	AP@0.5 (%)
Category I (unripe)	84.2	90.7	93.0
Category II (semi-ripe)	79.1	82.6	86.3
Category III (ripe)	93.4	95.6	98.0
Overall (mAP@0.5)	87.0	89.6	92.5

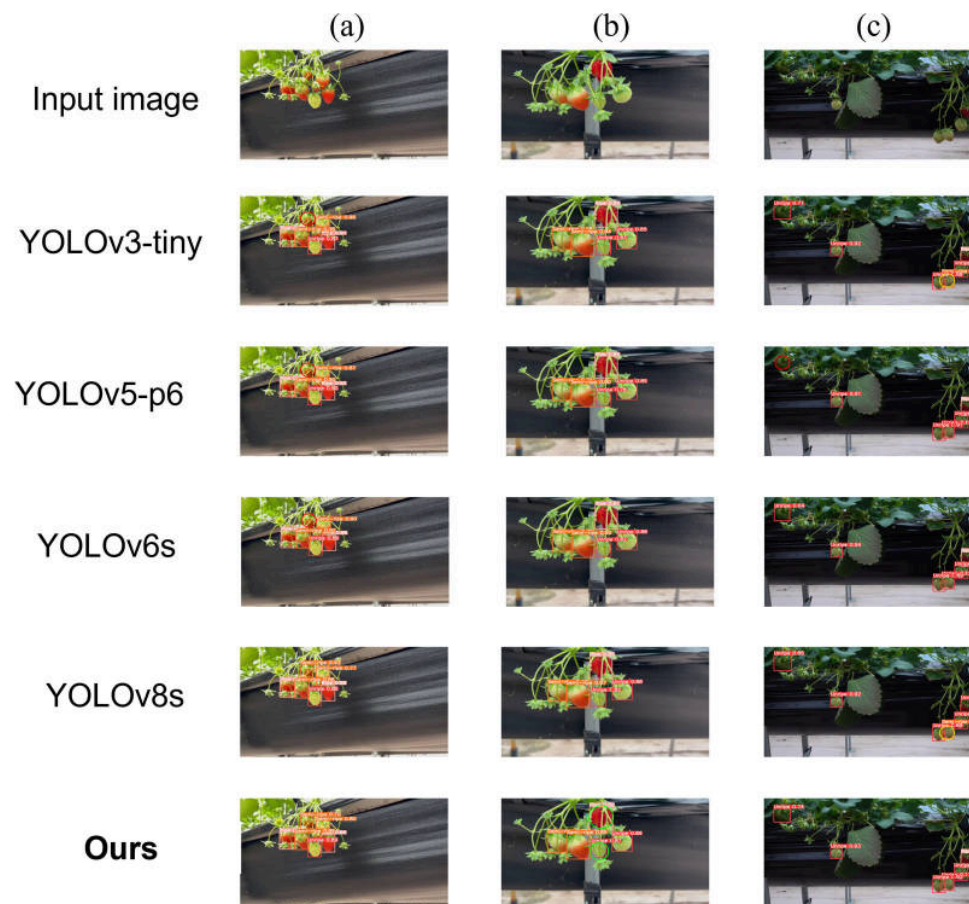


Figure 13. Qualitative comparison of detection results from different models on challenging greenhouse images. Test images represent common scenarios: (a) Dense clustering; (b) occlusions, and (c) Low light. Detection errors are explicitly marked: red circles indicate missed detections (false negatives), while yellow circles highlight misclassifications, and green circles denote instances where our demonstrated superior performance on occluded strawberries.

The model demonstrated excellent ability in identifying Category (III) ripe strawberries, achieving the highest AP of 98.0%. This result is likely due to the highly distinct and consistent visual features of this class. Ripe strawberries typically exhibit a uniform, deep-red coloration and a distinct shape that creates a strong chromatic and morphological contrast against the green foliage and black plastic mulch of the greenhouse background. This high degree of feature separation enables the model's convolutional layers to learn a robust and unambiguous representation, resulting in very few false positives or negatives. From a practical view, this high accuracy is critically important, as the correct identification of ripe fruit is the most valuable output for guiding immediate harvesting decisions and estimating market-ready yield.

The performance on Category I (unripe) fruits was also highly robust, with an AP of 93.0%. This indicates the model's strong effectiveness in distinguishing small green and white strawberries from the surrounding plant canopy. This task is non-trivial, as it involves overcoming the classic “green-on-green” detection problem common in agricultural computer vision [48,49]. In this case, the model likely learned to rely less on color contrast and more on morphological and textural cues, such as the characteristic shape and the pattern of achenes, for the downstream task of in-field fruit counting.

The most significant analytical result comes from the performance on Category II (semi-ripe), which was identified as the most challenging class for detection, with a lower AP of 86.3%. While this is still a strong result, the performance drop relative to the other

classes is noteworthy. This difficulty is attributable to two primary factors. The first is the inherent visual ambiguity of this transitional stage. Category II (semi-ripe) fruits lack a consistent look; they present a mottled appearance with a variable mixture of red, white, and green patches. This inconsistency makes it hard for the model to learn a single, coherent feature representation, leading to a higher chance of misclassification as either ripe or unripe. The second factor is the class imbalance within the training dataset. With significantly fewer annotated instances compared to the other two classes, the model had less data from which to learn the full range of variations within this category. This situation makes it more prone to error. This finding is consistent with the broader literature, where studies such as [17,32,33] have also reported similar results.

3.3. Ablation Study of Model Improvements

An ablation study was conducted to systematically isolate and measure the individual contributions of our three primary modifications: the replacement of the C2f module with the lightweight C3x module, the additional detection head, and the adoption of the WIoU loss function over the standard CIoU. To ensure a fair comparison, all model configurations were trained and evaluated under identical conditions using our custom dataset. The detailed outcomes of these empirical analyses are presented in Table 8.

Table 8. Results of the ablation study.

Model	C3x Module	Head	WIoU	I	AP (%) II	III	mAP@50 (%)	Parameters (M)
Baseline YOLOv8s (A)	×	×	×	92.5	86.1	97.9	92.1	11.1
B	✓	×	×	92.1	85.3	96.7	91.4	8.6
C	✓	✓	×	92.5	86.6	97.8	92.3	8.05
D	✓	✓	✓	93.0	86.3	98.0	92.5	8.0

The YOLOv8s model (Model A) served as the baseline, achieving a mAP@0.5 of 92.1% and a considerable parameter count of 11.1 million (M). The first modification involved integrating the lightweight C3x module in place of the standard backbone (Model B). This change produced a significant and immediate benefit in terms of computational efficiency. The parameter count was reduced by 22.5% to 8.6 M. Meanwhile, this improvement in efficiency was followed by a slight reduction in overall performance, with the mAP@0.5 decreasing by 0.7% to 91.4%. A closer examination of the per-class results reveals that this performance drop was consistent across all ripeness categories. This initial result confirms that while the C3x module is highly effective at reducing model complexity, its more generalized feature extraction capabilities come at a minor cost to specialized detection accuracy when implemented in isolation.

The next stage of the study introduced the additional detection head combined with the C3x module (Model C). This combination proved to be highly synergistic and was critical to the success of our model. Comparing Model C to Model B, the mAP@0.5 increased from 91.4% to 92.3%, not only recovering the performance lost from the backbone modification but surpassing the original baseline by 0.2%, all while further reducing the model's parameter count to 8.05 M. The per-class metrics provide a clear explanation for this improvement. The AP for the most challenging Category II (semi-ripe) strawberries rose significantly from 85.3% to 86.6%. This demonstrates that the new head, specifically designed to process higher-resolution features for small and ambiguous targets, effectively compensated for the architectural simplification of the backbone. It provided the specialized

capability needed to accurately detect small objects, compensating for any trade-offs in feature representation.

Our optimization strategy is represented by our final proposed model (Model D), which incorporates all three modifications. The final step involved replacing the standard loss function with WIoU. Comparing Model D with Model C, the addition of WIoU provided the final refinement, elevating the overall mAP@0.5 from 92.3% to its peak value of 92.5% and reducing the parameter count to 8 M. This enhancement was primarily driven by improved performance on the more visually distinct classes, with the AP for Category I (unripe) increasing from 92.5% to 93.0% and for Category III (ripe) increasing from 97.8% to 98.0%. This suggests that the WIoU loss function's dynamic weighting strategy was particularly effective at improving the model's localization accuracy for well-defined targets, enabling more precise bounding box regression.

These results validate that both the architectural and loss function improvements contributed positively to the model's final balance of high accuracy and computational efficiency.

3.4. Model Efficiency and Computational Cost

Beyond detection accuracy, the practical feasibility of deploying deep learning models in agricultural environments, especially on mobile or embedded systems, is strongly dependent on computational efficiency. A model that achieves high accuracy at the cost of excessive resource usage may be impractical for field deployment. To assess this trade-off, we conducted a comprehensive evaluation of our proposed model against the benchmark detectors, comparing metrics including parameter count, computational complexity (GFLOPs), model size, and inference speed. The results, summarized in Table 9, show that our model not only achieves high detection performance but also delivers substantial efficiency improvements.

Table 9. Comparative analysis of model efficiency and computational cost.

Model	Parameters (M)	GFLOPs	Model Size (MB)	Inference Time (ms)
YOLOv3 tiny	12.1	18.9	23.2	5.6
YOLOv5s-p6	15.3	24.2	29.4	10.6
YOLOv6s	16.3	44.0	31.3	11.0
YOLOv8s	11.1	28.4	21.4	8.9
YOLOv8m	25.8	78.7	49.6	17.2
YOLOv8n	3.0	10.1	9.94	6.8
Ours	8.0	27.1	15.7	9.7

The most notable strength of our model lies in its compact architecture. With only 8.0 M parameters, it is lightweight with 27.9% fewer parameters than YOLOv8s, nearly half that of YOLOv5s-p6 and YOLOv6s, and over three times smaller than the YOLOv8m. Conversely, while YOLOv8n is smaller (3.0 M parameters), our model achieves a substantial 2.0% absolute increase in mAP, a necessary trade-off for reliable performance in complex scenes. With a model size of only 15.7 MB, our model is suitable for deployment on devices with limited storage capacity, such as embedded systems or edge computing platforms.

In terms of computational complexity, our model runs at 27.1 GFLOPs. This is close to YOLOv8s (28.4 GFLOPs) while it is much more efficient than YOLOv6s (44.0 GFLOPs). This moderate complexity ensures lower energy consumption and processing load. Despite its reduced size and complexity, our model maintains a competitive inference speed of 9.7 ms per image, equivalent to approximately 103 FPS, which supports real-time video processing. While YOLOv3-tiny and YOLOv8n demonstrate faster speeds of 5.6 ms and 6.8 ms, it is at the expense of accuracy and architectural depth. Our model's speed also

outpaces YOLOv5s-p6 (10.6 ms) and YOLOv6s (11.0 ms), and is only slightly slower than YOLOv8s (8.9 ms), all while maintaining a more compact and efficient design.

Importantly, these efficiency increments do not come at the cost of performance. Compared to the benchmark models, our model achieved a high detection accuracy on the strawberry dataset (detailed in Section 3.1), demonstrating that our architectural improvements preserve and enhance detection quality. These results show that our optimization strategy has successfully produced a model that achieves an optimal balance of accuracy, size, and speed.

3.5. Counting Algorithm Evaluation

The practical efficiency of the complete framework, an automated counting system, integrating our improved YOLOv8s detector with the ByteTrack algorithm, was evaluated using the dedicated video dataset comprising 20 distinct streams captured under greenhouse conditions (as described in Section 2.3). The system's automated counts are compared against ground-truth counts. An overview of the system's counting workflow is provided in Figure 14, illustrating how unique track-IDs are assigned to individual strawberries as they cross the virtual counting line (ROI), with the total counts increasing as the frames progress.

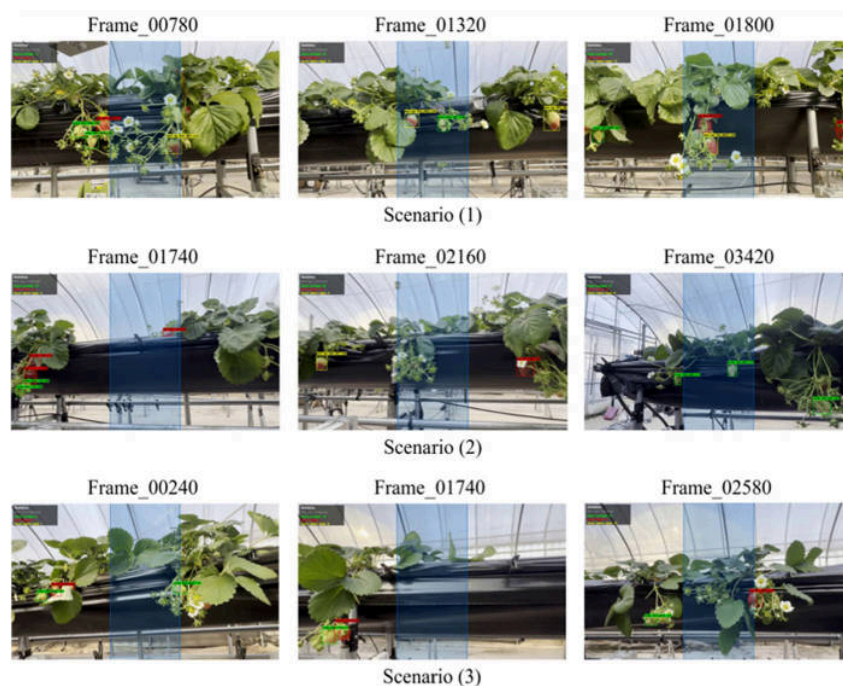


Figure 14. Sample Instances of the strawberry tracking and counting algorithm. The sequences show the successful assignment of track-IDs and the incrementing of class-specific counts as fruits pass the ROI, validating the core logic of the counting system.

The overall performance of the counting algorithm is quantitatively assessed using linear regression analysis, as shown in Figure 15. The scatter plot, Figure 15a, compares the total number of strawberries predicted by the system against the total ground-truth count for each video sequence. The analysis reveals a strong positive correlation. The system achieved a high coefficient of determination (R^2) of 0.914, indicating that 91.4% of the variance in the ground-truth counts is predictable from the model's output. This confirms a high degree of system-level consistency and reliability in yield estimation.

Further statistical evaluation was conducted to measure the absolute deviation between predicted and true counts. The system achieved a Mean Absolute Error (MAE) of 8.55 fruits, a Root Mean Squared Error (RMSE) of 9.63 fruits, and a Mean Absolute

Percentage Error (MAPE) of 9.52%. These values suggest that the system maintains an average prediction error of less than 10%, supporting its robustness for in-field conditions.

In addition to the aggregate analysis, a more detailed class evaluation was conducted on the system's capabilities for each ripeness class. The individual regression plots for each category are presented in Figure 15b–d, with their corresponding metrics summarized in Table 10. The system demonstrated its highest performance for Category III (ripe) strawberries, achieving an R^2 of 0.950 and the lowest MAE of 2.65 fruits. This result correlation represents the system's reliability for the most commercial task of estimating harvest-ready yield. Similarly, detection of unripe fruits was robust, with an R^2 of 0.910 and a MAPE of 9.62%, supporting the system's suitability for monitoring future yield potential.

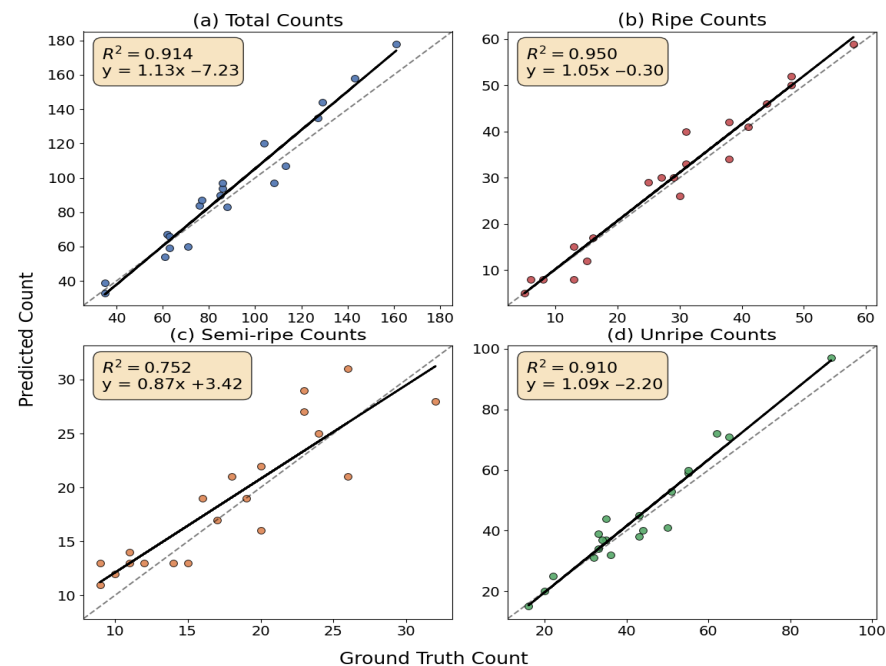


Figure 15. Strawberry fruit count results showing scatter plot of system's predicted vs. ground truth counts for (a) Total; (b) Ripe; (c) Semi-ripe; and (d) Unripe.

Table 10. Performance metrics for the automated counting system by ripeness category.

Ripeness Category	R^2	MAE	RMSE	MAPE
Category III (ripe)	0.950	2.65	3.37	11.63
Category II (semi-ripe)	0.752	2.70	3.16	16.25
Category I (unripe)	0.910	4.20	5.07	9.62
Total (overall)	0.914	8.55	9.63	9.52

Conversely, the semi-ripe category exhibited comparatively lower accuracy with an R^2 of 0.752 and the highest MAPE of 16.25%. This drop in performance aligns with detection-stage observations and is primarily attributed to the inherent visual ambiguity of this transitional ripeness stage. Misclassification errors, where semi-ripe fruits are occasionally confused with unripe or ripe ones, propagate into the tracking and counting pipeline, impacting class-specific tallies even when the total fruit count remains relatively stable.

To display the practical variability of the system's performance, the bar chart in Figure 16 provides a video analysis, illustrating the deviation between predicted and ground-truth counts for each of the twenty video streams. This visualization highlights instances of both slight under-counting (video 19, with an accuracy of 84.5%) and over-counting (video 4, with an accuracy of 89.4%). The individual video counting accuracies

ranged from a low of 84.51% to a high of 95.24%. These variations are directly linked to the different complexities within each video file, such as the density of fruit clusters, the severity of leaf occlusion, and fluctuations in ambient lighting, demonstrating the system's response to real-world challenges. Collectively, the quantitative and qualitative evaluations provide a comprehensive validation of our integrated counting system. The high overall R^2 value confirms its reliability for total yield estimation. At the same time, the detailed per-class analysis demonstrates its effectiveness as a tool for successfully identifying and classifying different strawberry classes.

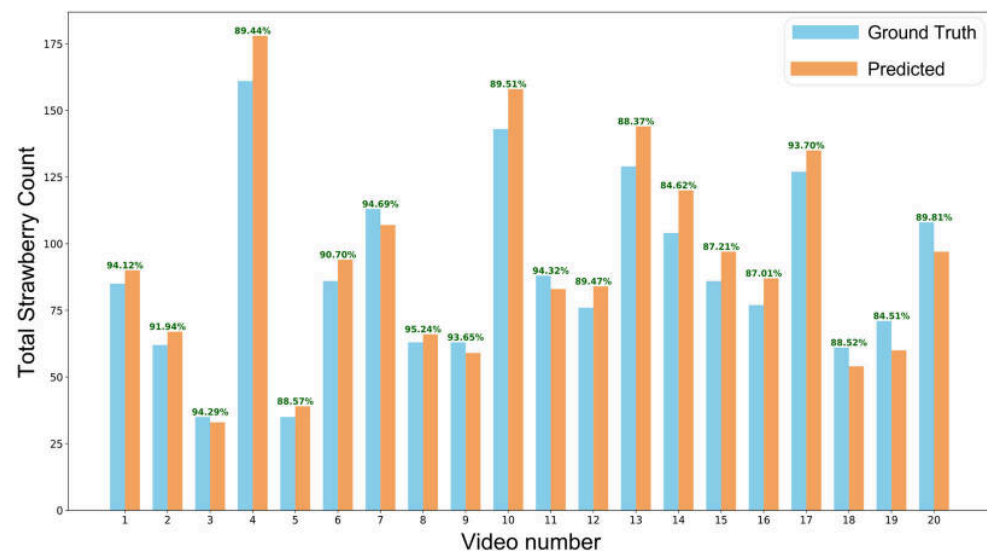


Figure 16. Comparison of predicted and ground-truth counts for each of the 20 videos, with individual counting accuracies shown.

3.6. Comparison with Other Counting Algorithms

A direct numerical comparison of counting results across different studies is inherently challenging due to differences in datasets, camera quality, and environmental conditions; however, a thorough discussion of our framework in the context of both the broader state-of-the-art research and a controlled internal benchmark is essential for validating our methodological choices. This section first positions our work relative to other recent, notable strawberry counting systems and then presents an empirical validation of our selected tracking component.

Our approach is best understood in comparison with other similar research work. For instance, the study by Li et al. [30] focused on maximizing detection accuracy by enhancing the two-stage Faster R-CNN architecture, achieving a high mAP of 87.3% and a counting accuracy of 99.1% on ripe fruit. The primary difference lies in our deliberate choice of a one-stage YOLOv8s architecture, which was aimed at achieving a superior balance between high accuracy and the real-time inference speeds crucial for video-based tracking. In the study by Zhou et al. [31], the critical challenge of occlusion was targeted by integrating YOLOv5s with a multiple mapping algorithm, achieving a low relative counting error of 6.7%. This highlights the importance of occlusion resilience, a problem we addressed through enhancing the detector's intrinsic localization capabilities with the WIoU loss function and leveraging ByteTrack's proven robustness to temporary object disappearances.

To empirically validate our choice of ByteTrack as the tracking algorithm, we conducted a controlled internal comparison. To achieve this, we tested the performance of our complete framework against systems using the same optimized YOLOv8s detector but replacing ByteTrack with four other widely used MOT algorithms: SORT, DeepSORT,

Norfair [50], and MOTPY [51]. This methodology ensures that any observed differences in counting accuracy are directly attributable to the tracking component itself.

The comprehensive results of this investigation are shown in Figure 17. The data demonstrates the superior performance of our proposed system, which pairs our detector with ByteTrack. This configuration achieved the highest coefficient of determination and the lowest errors, substantiating its robustness and reliability in the target environment.

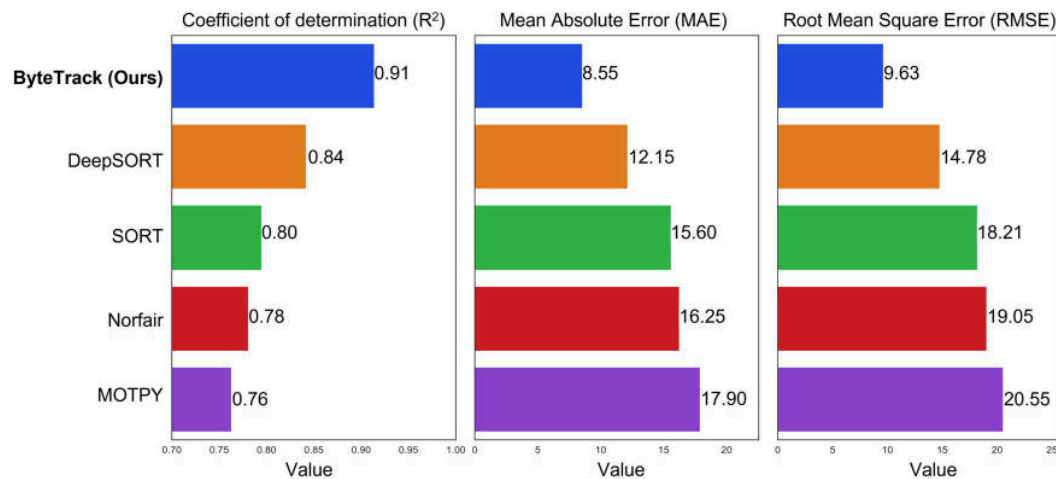


Figure 17. Comparative performance of different MOTs paired with the improved YOLOv8s detector.

In contrast, the alternative tracking algorithms struggled significantly to adapt to the specific challenges of the in-field strawberry environment. The DeepSORT algorithm, while powerful for tracking objects with distinct appearances such as pedestrians, proved not ideal for strawberries. The core issue lies in its reliance on a Re-ID module for appearance matching. The subtle visual similarity between individual fruits, compounded by slight changes in appearance due to lighting variations, led to frequent feature-matching errors, resulting in numerous misidentifications. This manifested as significant over-counting and a consequently lower R^2 of 0.842.

The trackers that rely more heavily on motion prediction, SORT, Norfair, and MOTPY, exhibited even greater performance degradation. These algorithms were particularly susceptible to tracking failures during occlusions. In the dense strawberry canopy, when a leaf temporarily hides a fruit, these methods often failed to re-establish the track upon reappearance, leading to fragmented trajectories and a high number of undercounts. The irregular, non-linear motion of the manually guided data collection platform further compounded these issues, as the simple Kalman filter motion predictions were often insufficient to bridge the gap during periods of occlusion.

3.7. Limitations and Future Studies

While the proposed framework demonstrates strong performance and practical utility, several limitations define its current scope and highlight directions for future research. Firstly, the dataset, though diverse in lighting conditions and growth stage, was collected from a single experimental greenhouse populated with three Korean strawberry cultivars (Seolhyang, Keumsil and Hongee). Given that strawberry morphology and environmental factors vary across cultivars and cultivation systems (open-field vs. greenhouse), the model's generalizability may be limited without further training and validation on broader datasets and different conditions. To address this and enhance the practical value of our framework, future work will focus on creating a comprehensive, multi-source dataset from different geographical locations and with a wider range of cultivars with distinct morphological traits, and varied cultivation systems, including open-field farms.

Secondly, the ripeness classification in this study was based on a three-stage discrete scale (unripe, semi-ripe, and ripe), which simplifies the continuous process of fruit maturation. While this framework offered practicality, this abstraction introduced classification challenges, particularly for the visually ambiguous semi-ripe category. This limitation suggests that representing a gradual biological process through only three classes reduces precision and model robustness. Future research should therefore consider more sophisticated ripeness evaluation methods, such as adopting a more granular classification scheme with additional intermediate stages or developing a maturity evaluation model based on continuous features (such as color histograms, spectral indices, or physiological indicators).

Thirdly, although the current framework provides an accurate count of fruit instances, it does not estimate yield in terms of biomass (weight). In practice, fruit size can vary considerably, meaning that a direct fruit count serves as an effective but indirect proxy for the actual harvestable weight, which is a more critical metric for commercial growers. Future work should aim to extend this framework to a comprehensive yield estimation system by integrating fruit size estimation to estimate yield in kilograms, this could be achieved by leveraging pixel dimensions of predicted bounding boxes and incorporating depth-sensing technologies.

4. Conclusions

This research presented a comprehensive, end-to-end framework for accurate and efficient ripeness-aware strawberry counting in complex greenhouse environments. Through the optimization of the YOLOv8s detector and integrating it with a robust multi-object tracking algorithm, this study successfully addressed key challenges related to small object detection, occlusion, and real-time computational performance.

The developed strawberry detection model, enhanced with a lightweight C3x module, an additional detection head, and the WIoU loss function, achieved an mAP@0.5 of 92.5%, outperforming the baseline YOLOv8s and other common variants while being 27.9% more lightweight. The complete counting system, which combined this optimized detector with the ByteTrack algorithm, demonstrated a high correlation with manual ground-truth counts, achieving an R^2 value of 0.914 and a low MAPE of 9.52%. The comparative analysis of trackers validated the selection of ByteTrack, which proved more resilient to the frequent, short-term occlusions in dense canopies than appearance-based DeepSORT or simpler motion-based SORT and Norfair trackers. Furthermore, the model's lightweight nature (8 M parameters) and model size (15.7 MB) confirms its suitability for real-time deployment on resource-constrained hardware, providing a robust foundation for the video-based counting pipeline.

In conclusion, this work successfully presents a practical and effective solution for automated yield estimation. The proposed framework strikes a balance between high accuracy and computational efficiency, providing a valuable tool for precision agriculture that facilitates more timely and informed management decisions.

Author Contributions: Conceptualization, Methodology, Formal analysis, Writing—Original Draft, Investigation, Data curation, Software, O.M.O.; Investigation, Resources, Writing—Review & Editing, S.K. and N.T.; Data curation, Validation, J.-H.K. and J.-I.C.; Investigation, Software, S.-M.K. and T.D.A.; Resources, Writing—Review & Editing, Supervision, Project administration, Funding acquisition, H.T.K. All authors have read and agreed to the published version of the manuscript.

Funding: This work was funded by Korea Institute of Planning and Evaluation for Technology in Food, Agriculture, Forestry and Fisheries (IPET) through Agriculture, Food and Rural Affairs Convergence Technologies Program for Educating Creative Global Leader, funded by Ministry of Agriculture, Food and Rural Affairs (MAFRA) (717001-7). This work was supported by SMART FARM Research Center in Gyeongsang National University (GNU), funded by Sacheon-si.

Data Availability Statement: The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Acknowledgments: This work was funded by Korea Institute of Planning and Evaluation for Technology in Food, Agriculture, Forestry and Fisheries (IPET) through Agriculture, Food and Rural Affairs Convergence Technologies Program for Educating Creative Global Leader, funded by Ministry of Agriculture, Food and Rural Affairs (MAFRA) (717001-7). This work was supported by SMART FARM Research Center in Gyeongsang National University (GNU), funded by Sacheon-si.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Shin, S.H.; Deb, N.C.; Arulmozhi, E.; Tamrakar, N.; Ogundele, O.M.; Kook, J.; Kim, D.H.; Kim, H.T. Prediction of Carbon Dioxide Concentrations in Strawberry Greenhouse by Using Time Series Models. *Agriculture* **2024**, *14*, 1895. [CrossRef]
2. Shi, Y.; Shen, Y.; Bai, J.; Ahmad, B.; Rahman, M.U.; Chen, Q.; Wen, Z. Detached Leaf Assay-Based Screening of Anthracnose Resistant Wild Strawberry Genotypes. *Trop. Plant Pathol.* **2024**, *49*, 876–885. [CrossRef]
3. Ullah, I.; Demirsoy, H.; Soysal, D.; Lizalo, A.; Doğan, D.E.; Demirsoy, L. Evaluation of Strawberry Cultivars Based on Growth-Related Attributes. *Appl. Fruit Sci.* **2024**, *66*, 431–439. [CrossRef]
4. Statistics Korea. Production Area and Volume of Vegetable in South Korea in 2022. Available online: <https://kostat.go.kr/anse/> (accessed on 19 July 2025).
5. Han, H.; Jang, Y.J.; Han, K.; Park, H.-N.; Kim, D.-S.; Lee, S.; Oh, Y. Chromosome-Level Genome Assembly of Cultivated Strawberry ‘Seolhyang’ (*Fragaria × ananassa*). *Sci. Data* **2025**, *12*, 1002. [CrossRef]
6. Chai, J.; Zeng, H.; Li, A.; Ngai, E.W.T. Deep Learning in Computer Vision: A Critical Review of Emerging Techniques and Application Scenarios. *Mach. Learn. Appl.* **2021**, *6*, 100134. [CrossRef]
7. Wang, A.; Qian, W.; Li, A.; Xu, Y.; Hu, J.; Xie, Y.; Zhang, L. NVW-YOLOv8s: An Improved YOLOv8s Network for Real-Time Detection and Segmentation of Tomato Fruits at Different Ripeness Stages. *Comput. Electron. Agric.* **2024**, *219*, 108833. [CrossRef]
8. Xiong, Y.; Ge, Y.; Grimstad, L.; From, P.J. An Autonomous Strawberry-harvesting Robot: Design, Development, Integration, and Field Evaluation. *J. Field Robot.* **2020**, *37*, 202–224. [CrossRef]
9. Hu, H.; Kaizu, Y.; Zhang, H.; Xu, Y.; Imou, K.; Li, M.; Huang, J.; Dai, S. Recognition and Localization of Strawberries from 3D Binocular Cameras for a Strawberry Picking Robot Using Coupled YOLO/Mask R-CNN. *Int. J. Agric. Biol. Eng.* **2022**, *15*, 175–179. [CrossRef]
10. Zhang, D.; Lee, D.-J.; Tippetts, B.J.; Lillywhite, K.D. Date Maturity and Quality Evaluation Using Color Distribution Analysis and Back Projection. *J. Food Eng.* **2014**, *131*, 161–169. [CrossRef]
11. Wang, C.; Tang, Y.; Zou, X.; Luo, L.; Chen, X. Recognition and Matching of Clustered Mature Litchi Fruits Using Binocular Charge-Coupled Device (CCD) Color Cameras. *Sensors* **2017**, *17*, 2564. [CrossRef]
12. Karki, S.; Basak, J.K.; Paudel, B.; Deb, N.C.; Kim, N.-E.; Kook, J.; Kang, M.Y.; Kim, H.T. Classification of Strawberry Ripeness Stages Using Machine Learning Algorithms and Colour Spaces. *Hortic. Environ. Biotechnol.* **2024**, *65*, 337–354. [CrossRef]
13. Karki, S.; Tamrakar, N.; Kook, J.H.; Ogundele, O.M.; Basak, J.K.; Kang, M.Y.; Kim, H.T. On-Field Grading of Strawberries in a Greenhouse Environment with Deep Learning-Based Occlusion Handling. *Hortic. Environ. Biotechnol.* **2025**, *43*, 1–17. [CrossRef]
14. Redmon, J.; Divvala, S.; Girshick, R.; Farhadi, A. You Only Look Once: Unified, Real-Time Object Detection. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Las Vegas, NV, USA, 27–30 June 2016; pp. 779–788.
15. Ren, S.; He, K.; Girshick, R.; Sun, J. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. *Adv. Neural Inf. Process. Syst.* **2015**, *28*, 91–99. [CrossRef] [PubMed]
16. Sirisha, U.; Praveen, S.P.; Srinivasu, P.N.; Barsocchi, P.; Bhoi, A.K. Statistical Analysis of Design Aspects of Various YOLO-Based Deep Learning Models for Object Detection. *Int. J. Comput. Intell. Syst.* **2023**, *16*, 126. [CrossRef]
17. An, Q.; Wang, K.; Li, Z.; Song, C.; Tang, X.; Song, J. Real-Time Monitoring Method of Strawberry Fruit Growth State Based on YOLO Improved Model. *IEEE Access* **2022**, *10*, 124363–124372. [CrossRef]
18. He, Z.; Karkee, M.; Upadhayay, P. Detection of Strawberries with Varying Maturity Levels for Robotic Harvesting Using YOLOv4. In Proceedings of the 2021 ASABE Annual International Virtual Meeting, Online, 12–16 July 2021; American Society of Agricultural and Biological Engineers: St. Joseph, MI, USA, 2021; p. 1.
19. Wang, D.; Wang, X.; Chen, Y.; Wu, Y.; Zhang, X. Strawberry Ripeness Classification Method in Facility Environment Based on Red Color Ratio of Fruit Rind. *Comput. Electron. Agric.* **2023**, *214*, 108313. [CrossRef]
20. Zhang, Y.; Liang, Y.; Leng, J.; Wang, Z. SCGTracker: Spatio-Temporal Correlation and Graph Neural Networks for Multiple Object Tracking. *Pattern Recognit.* **2024**, *149*, 110249. [CrossRef]
21. Bewley, A.; Ge, Z.; Ott, L.; Ramos, F.; Upcroft, B. Simple Online and Realtime Tracking. In Proceedings of the 2016 IEEE International Conference on Image Processing (ICIP), Phoenix, AZ, USA, 25–28 September 2016; pp. 3464–3468.

22. Wojke, N.; Bewley, A.; Paulus, D. Simple Online and Realtime Tracking with a Deep Association Metric. In Proceedings of the 2017 IEEE International Conference on Image Processing (ICIP), Beijing, China, 17–20 September 2017; pp. 3645–3649.
23. Zhang, Y.; Sun, P.; Jiang, Y.; Yu, D.; Weng, F.; Yuan, Z.; Luo, P.; Liu, W.; Wang, X. Bytetrack: Multi-Object Tracking by Associating Every Detection Box. In Proceedings of the European Conference on Computer Vision, Tel Aviv, Israel, 23–27 October 2022; Springer: Cham, Switzerland, 2022; pp. 1–21.
24. Wen, L.; Du, D.; Cai, Z.; Lei, Z.; Chang, M.-C.; Qi, H.; Lim, J.; Yang, M.-H.; Lyu, S. UA-DETRAC: A New Benchmark and Protocol for Multi-Object Detection and Tracking. *Comput. Vision. Image Underst.* **2020**, *193*, 102907. [[CrossRef](#)]
25. Razzok, M.; Badri, A.; El Mourabit, I.; Ruichek, Y.; Sahel, A. Pedestrian Detection and Tracking System Based on Deep-SORT, YOLOv5, and New Data Association Metrics. *Information* **2023**, *14*, 218. [[CrossRef](#)]
26. Niu, L.; Fan, Y.; Liu, T.; Han, Q. Momt: A Maritime Real-Time Visual Multi-Object Tracking Algorithm Based on Unmanned Surface Vehicles. *IEEE Sens. J.* **2024**, *12*, 35429–35447. [[CrossRef](#)]
27. Gao, F.; Fang, W.; Sun, X.; Wu, Z.; Zhao, G.; Li, G.; Li, R.; Fu, L.; Zhang, Q. A Novel Apple Fruit Detection and Counting Methodology Based on Deep Learning and Trunk Tracking in Modern Orchard. *Comput. Electron. Agric.* **2022**, *197*, 107000. [[CrossRef](#)]
28. He, L.; Wu, F.; Du, X.; Zhang, G. Cascade-SORT: A Robust Fruit Counting Approach Using Multiple Features Cascade Matching. *Comput. Electron. Agric.* **2022**, *200*, 107223. [[CrossRef](#)]
29. Lin, Y.; Chen, T.; Liu, S.; Cai, Y.; Shi, H.; Zheng, D.; Lan, Y.; Yue, X.; Zhang, L. Quick and Accurate Monitoring Peanut Seedlings Emergence Rate through UAV Video and Deep Learning. *Comput. Electron. Agric.* **2022**, *197*, 106938. [[CrossRef](#)]
30. Li, J.; Zhu, Z.; Liu, H.; Su, Y.; Deng, L. Strawberry R-CNN: Recognition and Counting Model of Strawberry Based on Improved Faster R-CNN. *Ecol. Inform.* **2023**, *77*, 102210. [[CrossRef](#)]
31. Zhou, X.; Zhang, Y.; Jiang, X.; Riaz, K.; Rosenbaum, P.; Lefsrud, M.; Sun, S. Advancing Tracking-by-Detection with MultiMap: Towards Occlusion-Resilient Online Multiclass Strawberry Counting. *Expert. Syst. Appl.* **2024**, *255*, 124587. [[CrossRef](#)]
32. Wang, Y.; Yan, G.; Meng, Q.; Yao, T.; Han, J.; Zhang, B. DSE-YOLO: Detail Semantics Enhancement YOLO for Multi-Stage Strawberry Detection. *Comput. Electron. Agric.* **2022**, *198*, 107057. [[CrossRef](#)]
33. Tamrakar, N.; Karki, S.; Kang, M.Y.; Deb, N.C.; Arulmozhi, E.; Kang, D.Y.; Kook, J.; Kim, H.T. Lightweight Improved YOLOv5s-CGHOSTNET for Detection of Strawberry Maturity Levels and Counting. *AgriEngineering* **2024**, *6*, 962–978. [[CrossRef](#)]
34. Tamrakar, N.; Paudel, B.; Karki, S.; Deb, N.C.; Arulmozhi, E.; Kook, J.H.; Kang, M.Y.; Kang, D.Y.; Ogundele, O.M.; Nakarmi, B.; et al. Peduncle Detection of Ripe Strawberry to Localize Picking Point Using DF-Mask R-CNN and Monocular Depth. *IEEE Access* **2025**, *13*, 73889–73902. [[CrossRef](#)]
35. Yang, G.; Wang, J.; Nie, Z.; Yang, H.; Yu, S. A Lightweight YOLOv8 Tomato Detection Algorithm Combining Feature Enhancement and Attention. *Agronomy* **2023**, *13*, 1824. [[CrossRef](#)]
36. Khan, Z.; Liu, H.; Shen, Y.; Zeng, X. Deep Learning Improved YOLOv8 Algorithm: Real-Time Precise Instance Segmentation of Crown Region Orchard Canopies in Natural Environment. *Comput. Electron. Agric.* **2024**, *224*, 109168. [[CrossRef](#)]
37. Wang, T.; Ren, S.; Zhang, H. Nighttime Wildlife Object Detection Based on YOLOv8-night. *Electron. Lett.* **2024**, *60*, e13305. [[CrossRef](#)]
38. Wu, T.; Dong, Y. YOLO-SE: Improved YOLOv8 for Remote Sensing Object Detection and Recognition. *Appl. Sci.* **2023**, *13*, 12977. [[CrossRef](#)]
39. Bochkovskiy, A.; Wang, C.-Y.; Liao, H.-Y.M. Yolov4: Optimal Speed and Accuracy of Object Detection. *arXiv* **2020**, arXiv:2004.10934. [[CrossRef](#)]
40. Liu, S.; Qi, L.; Qin, H.; Shi, J.; Jia, J. Path Aggregation Network for Instance Segmentation. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Salt Lake City, UT, USA, 18–23 June 2018; pp. 8759–8768.
41. Lin, T.-Y.; Maire, M.; Belongie, S.; Hays, J.; Perona, P.; Ramanan, D.; Dollár, P.; Zitnick, C.L. Microsoft Coco: Common Objects in Context. In Proceedings of the Computer Vision—ECCV 2014: 13th European conference, Zurich, Switzerland, 6–12 September 2014; Proceedings, Part v 13; Springer: Cham, Switzerland, 2014; pp. 740–755.
42. Cheng, Y.; Feng, G.; Zhang, C. An Efficient and Lightweight YOLOv8s Strawberry Maturity Detection Model. *J. Agric. Sci. Technol. A* **2024**, *14*, 46–66. [[CrossRef](#)]
43. Gao, S.; Cui, G.; Wang, Q. WCS-YOLOv8s: Improved YOLOv8s Model for Target Identification and Localisation throughout the Strawberry Growth Process. *Front. Plant Sci.* **2025**, *16*, 1579335. [[CrossRef](#)]
44. Li, J.; Li, J.; Zhao, X.; Su, X.; Wu, W. Lightweight Detection Networks for Tea Bud on Complex Agricultural Environment via Improved YOLO V4. *Comput. Electron. Agric.* **2023**, *211*, 107955. [[CrossRef](#)]
45. Zheng, Z.; Wang, P.; Ren, D.; Liu, W.; Ye, R.; Hu, Q.; Zuo, W. Enhancing Geometric Factors in Model Learning and Inference for Object Detection and Instance Segmentation. *IEEE Trans. Cybern.* **2021**, *52*, 8574–8586. [[CrossRef](#)]
46. Tong, Z.; Chen, Y.; Xu, Z.; Yu, R. Wise-IoU: Bounding Box Regression Loss with Dynamic Focusing Mechanism. *arXiv* **2023**, arXiv:2301.10051.

47. Prajwal, G.K.; Sanjaya, D.S.; Wandkar, S.M.; Basawaraj. Real-Time Multi-Object Detection and Tracking on Zynq UltraScale+ MPSoc. In Proceedings of the International Conference on Information and Communication Technology for Intelligent Systems, Las Vegas, LA, USA, 24 May 2022; Springer: Singapore, 2024; pp. 71–81.
48. Fu, X.; Zhao, S.; Wang, C.; Tang, X.; Tao, D.; Li, G.; Jiao, L.; Dong, D. Green Fruit Detection with a Small Dataset under a Similar Color Background Based on the Improved YOLOv5-AT. *Foods* **2024**, *13*, 1060. [[CrossRef](#)]
49. Wang, J.; Shang, Y.; Zheng, X.; Zhou, P.; Li, S.; Wang, H. Green Fruit Detector: Lightweight Green Fruit Detector in Orchard Environment. *PLoS ONE* **2024**, *19*, e0312164. [[CrossRef](#)]
50. Alori, J.; Descoins, A.; Javier; Lezama, F.; Yuhara, K.; Fernández, D.; Castro, A.; Fatih, D.; David; Cruz Linares, R.; et al. *Tryolabs/Norfair*, V2.2.0; Zenodo: Geneva, Switzerland, 2023. [[CrossRef](#)]
51. Wiktor, M. *Motpy—Simple Multi Object Tracking Library*, Version 0.0.10; GitHub: San Francisco, CA, USA, 2021. Available online: <https://github.com/wmuron/motpy> (accessed on 20 July 2025).

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.